There are two other points to note here. The NOT gate is a special case, having only one input. The second point is that a NAND gate is a combination of a NAND gate followed by a NOT gate, and a NOR gate is a combination of an OR gate followed by a NOT gate. NAND and NOR gates produce a complementary output to the AND and OR gates.

TASK 4.03

Draw a circuit where A and B are input to an AND gate, from which the output is carried to a NOT gate, from which there is an output X. Show that this has the same outcome as having one NAND gate.

Extension Question 4.01

Could the same outcome be produced by positioning a NOT gate before the AND gate?

TIP

You need to remember the symbol for each of these gates. A good start here is to remember that AN**D** has the proper **D** symbol and OR has the curvy one.

You also need to remember the definitions for the gates so that you can construct the corresponding truth table for each gate.

Question 4.01

Can you recall from memory the symbols and definitions of the six logic gates introduced in this chapter?

WORKED EXAMPLE 4.02

Constructing a logic circuit from a problem statement or logic expression

You need to be able to construct a logic circuit from either a problem statement or from a logic expression. If you are given a problem statement, the best approach is to first convert it to a logic expression.

Consider the following problem statement: A bank offers a special lending rate to customers subject to certain conditions. To qualify, a customer must satisfy certain criteria.

- The customer has been with the bank for two years.
- Two of the following conditions must also apply:
  - the customer is married
  - the customer is aged 25 years or older
  - the customer's parents are customers of the bank.

To convert this statement to a logic expression using symbols we can choose:

- let A represent an account held for two years
- let B represent that the customer is married
- let C represent that the customer is aged 25 years or older
- let D represent that the customer's parents have an account.

(Continued)

The logic expression can then be written as:

A AND (((B AND C) OR (B AND D)) OR (C AND D))

This could alternatively be presented with an outcome:

Special_rate IF A AND (((B AND C) OR (B AND D)) OR (C AND D))

alternatively as

X = A AND (((B AND C) OR (B AND D)) OR (C AND D))

Note the use of brackets to ensure that the meaning is clear. You may think that not all of the brackets are needed. In this example, an extra pair has been included to guide the construction of the circuit where only two inputs are allowed for any of the gates.

From this, we can see that the logic circuit corresponding to this logic expression derived from the original problem statement could be constructed using four AND gates and two OR gates as shown in Figure 4.03.

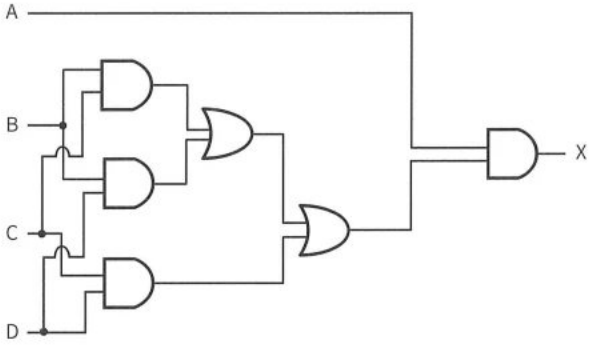


Figure 4.03 A logic circuit constructed from a problem statement

WORKED EXAMPLE 4.03

Constructing a truth table from a logic expression or logic circuit

You also need to be able to construct a truth table from either a logic expression or a logic circuit. We might have continued with the problem in Worked Example 4.02 but four inputs will lead to 16 rows in the truth table. Instead, we consider a slightly simpler problem with only three inputs and therefore only eight rows in the truth table. We will start with the circuit shown in Figure 4.04.

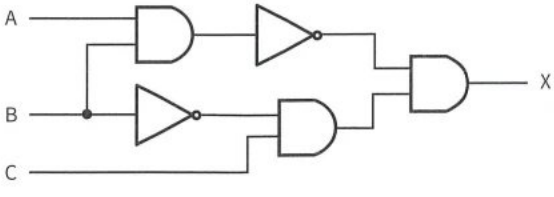


Figure 4.04 A circuit with three inputs for conversion to a truth table

Table 4.02 shows how the truth table needs to be set up initially. There are two points to note here. The first is that you must take care to include all of the eight different possible combinations of the input values. The second point is that for such a circuit it is not sensible to try to work out the outputs directly from the input values. Instead a systematic

(Continued)

approach should be used. This involves identifying intermediate points in the circuit and recording the values at each of them in the columns headed 'Workspace' in Table 4.02.

Inputs			Workspace				Output
A	B	C					X
0	0	0					
0	0	1					
0	1	0					
0	1	1					
1	0	0					
1	0	1					
1	1	0					
1	1	1					

Table 4.02 The initial empty truth table

Figure 4.05 shows the same circuit but with four intermediate points labelled M, N, P and Q identified. Each one has been inserted on the output side of a logic gate.

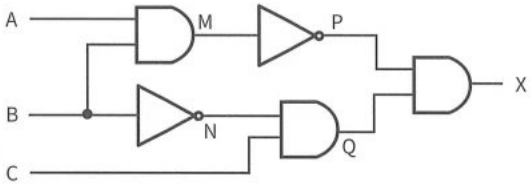


Figure 4.05 The circuit in Figure 4.04 with intermediate points identified

Now you need to work systematically through the intermediate points. You start by filling in the columns for M and N. Then you fill in the columns for P and Q which feed into the final AND gate. The final truth table is shown as Table 4.03. The circuit has two combinations of inputs that lead to a TRUE output from the circuit.

The columns containing the intermediate values (the workspace) could be deleted at this stage.

Inputs			Workspace				Output
A	B	C	M	N	P	Q	X
0	0	0	0	1	1	0	0
0	0	1	0	1	1	1	1
0	1	0	0	0	1	0	0
0	1	1	0	0	1	0	0
1	0	0	0	1	1	0	0
1	0	1	0	1	1	1	1
1	1	0	1	0	0	0	0
1	1	1	1	0	0	0	0

Table 4.03 The truth table for the circuit shown in Figure 4.05

One final point to make here is that you may be able to check part of your final solution by looking at just part of the circuit. For this example, if you look at the circuit you will see that the path from input C to the output passes through two AND gates. It follows, therefore, that for all combinations with C having value 0 the output must be 0. Therefore, in order to check your final solution you only need to examine the other four combinations of input values where C has value 1.

If a logic circuit is to be constructed from a truth table, the first stage is to create a logic expression. To do this only the rows producing a 1 output are used. Consider the truth table shown in Table 4.04. There are three rows producing a 1 output. Each of these produces a logic expression with AND operators. These three logic expressions are then combined with OR operators.

Inputs			Output
A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

Table 4.04 A truth table to be converted to a logic circuit

The three rows that produce a 1 output have the following values for the inputs:

A = 0, B = 0 and C = 1

A = 0, B = 1 and C = 1

A = 1, B = 0 and C = 0

Each one can be converted to a logical expression:

NOT A AND NOT B AND C

NOT A AND B AND C

A AND NOT B AND NOT C

The combination of the three such rows produces the following expression:

NOT A AND NOT B AND C

OR

NOT A AND B AND C

OR

A AND NOT B AND NOT C

This could be used to create a logic circuit, but the circuit would be quite complex. In Chapter 19 methods will be discussed that allow the simplest possible circuit to be constructed for a given logic problem.

If a logic expression is to be constructed from a logic circuit the first step is to construct a truth table from the circuit. Then the above method can be applied to this truth table.

TASK 4.04

An oven has a number of components that should all be working properly. For each component there is a signalling mechanism that informs a management system either if all is well, or if there is a problem. Table 4.05 summarises the signal values that record the status for each component.

Signal	Value	Component condition
A	0	Fan not working
	1	Fan working properly
B	0	Internal light not working
	1	Internal light working properly
C	0	Thermometer reading too high
	1	Thermometer reading in range

Table 4.05 Signals from the oven components

If the thermometer reading is in range but either or both the fan and light are not working, the management system has to output a signal to activate a warning light on the control panel. Draw a logic circuit for this fault condition.

Reflection Point:

Looking back over the chapter content, what would you say is the central concept in the subject matter?

Summary

- A logic scenario can be described by a problem statement or a logic expression.
- A logic expression comprises logic propositions and Boolean operators.
- Logic circuits are constructed from logic gates.
- The operation of a logic gate matches that of a Boolean operator.
- The outcome of a logic expression or a logic circuit can be expressed as a truth table.
- A logic expression can be created from a truth table using the rows that provide a 1 output.